

UrbanEats Expansion Analysis

Descriptive Analysis for Board Consultation – By Satvik Kumar

MIS41150 – Introduction to Business Analytics | October 2025

Executive Summary

The customer demand data analysis (January 2021 - September 2025) shows that there is a severe issue with operational efficiency: Branch D2 serves 27% of its customer base in remote districts (D12, D22, D20), which puts a significant operational burden on the branch despite high revenue performance. The southwestern corridor (Districts 12, 22, 24) has 235,767 customers but no local service, which means that customers have to travel and overburden the current branch operations.

Key Findings: Despite having the largest market share (42.6%), D2's dominant market position creates operational complexity; it is serving 95,233 customers (27%) in remote districts, which may result in high operating costs, longer delivery times, and a subpar customer experience. With successful operations and a variety of local sources of demand, D14 has a 30.9% share.

Implications for Network Planning: The data shows definite trends of operational inefficiency and unmet areas of demand. Districts 12, 22 and 24 show 235,767 customers and no effective coverage of the local branches and D2 is burdened with 70,633 customers in these remote locations. The evidence base of future expansion and network optimization decisions is provided with these findings.

1. Branch Performance Analysis

1.1 Overall Performance

Branch D2 is leading with 352,343 total customers and 618 average monthly demand, continuing to grow by 3.3% in 2025. D14 levels off at 255,593 customers (448 monthly average) with flat growth. D10 performs poorly with 218,655 customers (384 monthly average) and a -25.5% drop.

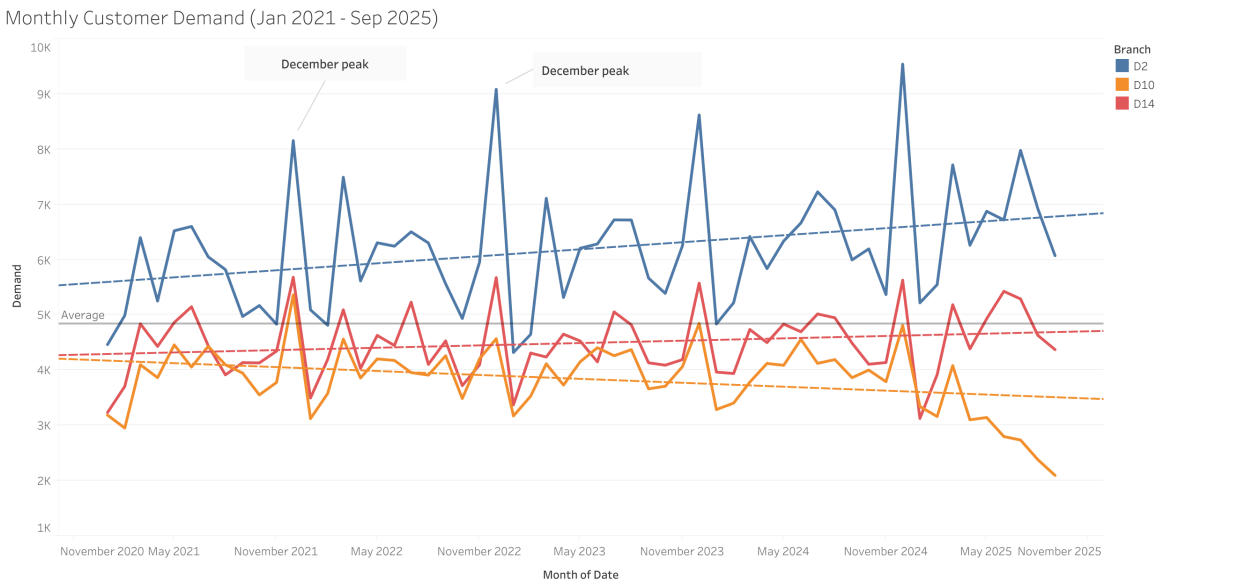
Branch	Total Customers	Monthly Avg	Market Share	2025 Trend
D2	352,343	618	42.6%	+3.3%
D14	255,593	448	30.9%	0%
D10	218,655	384	26.5%	-25.5%

Growth trends calculated by comparing Jan-Sep 2025 average monthly demand to Jan-Sep 2024 average.

1.2 D2 Operational Efficiency Issue

High Revenue Does Not Equal Efficient Operations: D2 has the highest revenue, but analysis shows that operations are highly inefficient. D2 has 95,233 customers (27.0% of its base) in remote districts (D12: 49,554; D20: 31,655; D22: 14,024). These customers require more delivery/travel time, which increases the per-customer costs, strain on kitchen/staff capacity, and poor customer experience to both long-distance travelers and local D2 residents. The core catchment areas of D2 (8, 4, 6, 6w) constitute just 67.9% of its clientele; the rest of the demand poses operational strain.

Seasonal Patterns: The average number of customers per month in December is 646 (33.3 more than the average number of customers per year 485), and this further compounds the pressure on the operations in the peak seasons. January has the lowest performance of 382 customers.



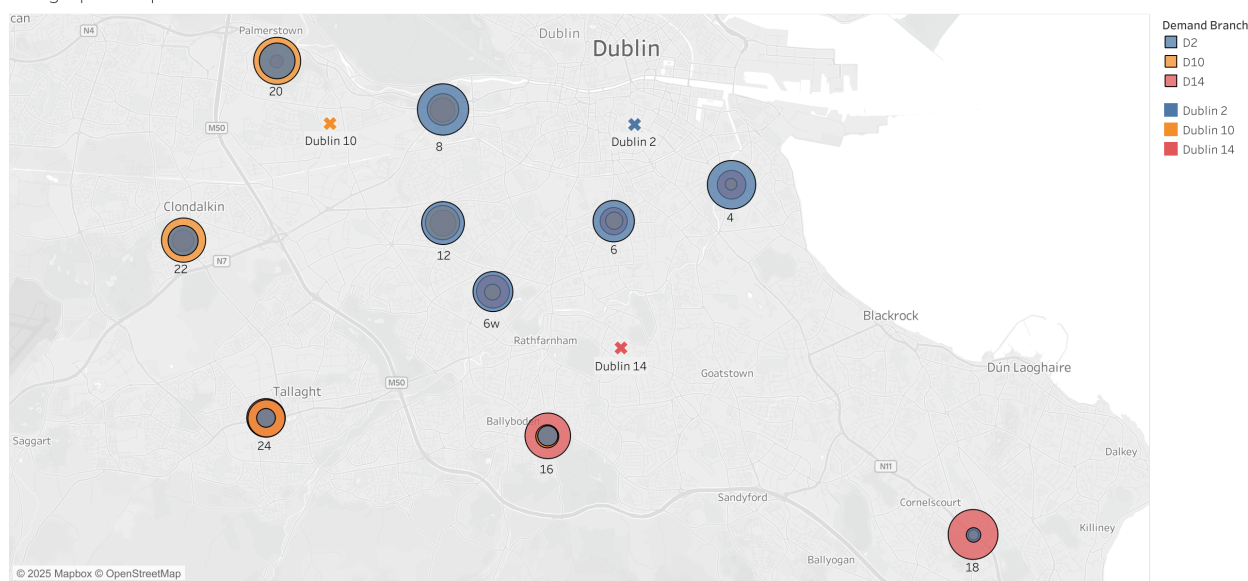
2. Geographic Demand Distribution

2.1 Demand Patterns at the District Level

District 8 has the greatest demand (107,963 customers), with D2 capturing 72.2%. District 20 is second (100,024 customers), with D10 capturing 65.8% of it. The five leading districts (8, 20, 4, 18, 12) represent 55.3% of the market.

District	Total Demand	Primary Branch	Primary %	D2 / D10 / D14
8	107,963	D2	72.2%	77,981 / 19,935 / 10,047
20	100,024	D10	65.8%	31,655 / 65,797 / 2,572
4	85,388	D2	82.4%	70,397 / 2,258 / 12,733
18	82,313	D14	93.0%	3,540 / 2,237 / 76,536
12	81,419	D2	60.9%	49,554 / 21,680 / 10,185
16	80,132	D14	80.3%	7,151 / 8,645 / 64,336

Geographic Map



2.2 Operational Efficiency: Overlapping vs. Distinct Demand

D2-D10 Overlapping Demand is Inefficient: D10's core inefficiency is its failure to capture its local market. The data clearly shows that despite District 8 being D10's closest neighbor, D2 captures 72.2% of its 107,963 customers. This "customer bypassing" is one of the network's problems: D10 resources appear underutilized, while D2 is servicing this distant demand, adding to its already high burden.

D14 Effective Operations with Separate Demand: D14 is operationally efficient since it is used in separate sources of local demand. District 18 has 93.0% dominance (76,536 customers) and low D2/D10 competition. District 16 is also efficient, with D14 having 80.3% (64,336 customers). This geographic focus allows D14 to simplify the operations, reduce the delivery time, and maintain the same quality of services, which proves the efficiency of serving the limited area of the local catchment.

2.3 Southwestern Corridor: Operational Strain of D2

Districts 12, 22, and 24 have 235,767 customers but no local branches, which means that customers have to drive to D2, D10, or D14. District 12: D2 takes 60.9% (49,554 customers) - 14.1% of the total base, and huge operation burden of D2. District 22: D10 tops with 71.4% (54,379 customers) - D10 has the highest source of demand but poses a delivery inefficiency problem. D24: Split coverage (D14 49%, D10 42%, D2 9%) shows that there is no branch that is effectively covering this area.

3. Key Insights and Business Implications

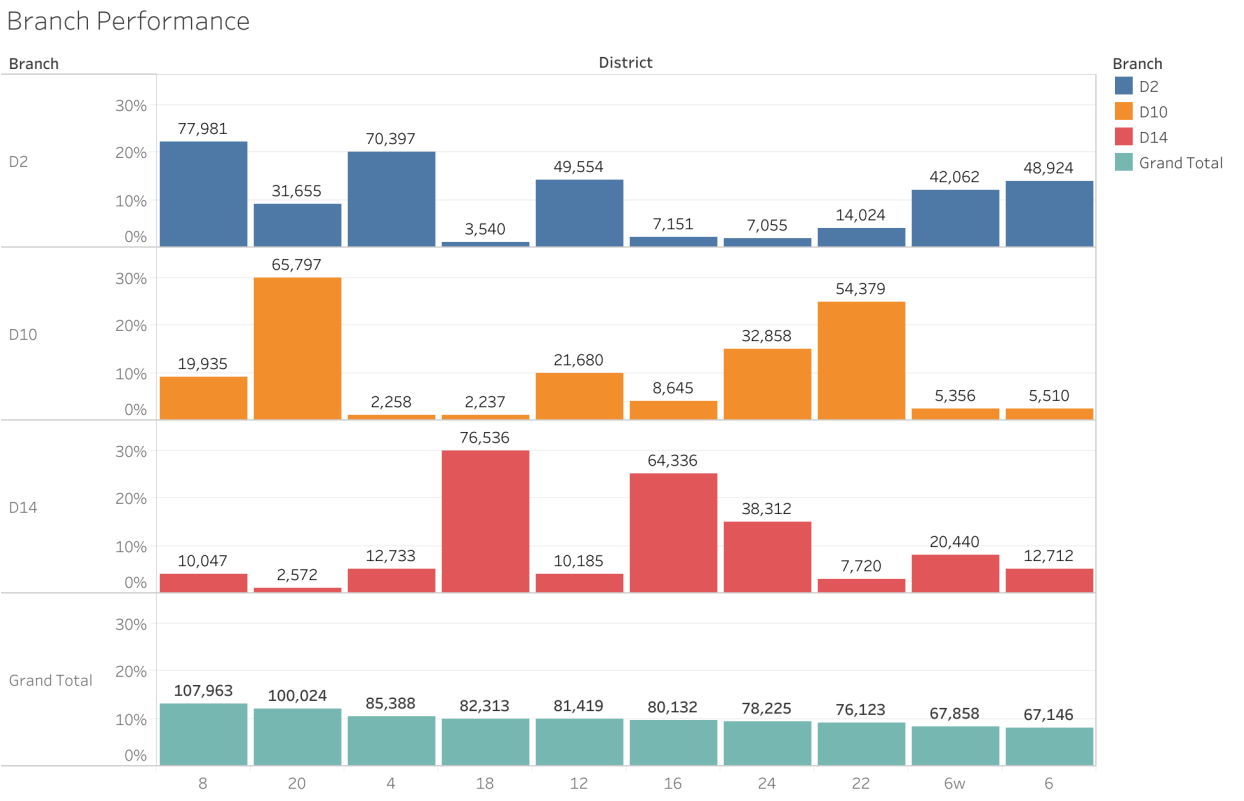
3.1 Network Efficiency Patterns

The comparison shows that there are unique operational efficiency trends in the three branches. The most efficient model is D14, which is a focused local catchment with a low level of competition. Conversely, both D2 and D10 lack efficiency because of overlapping sources of demand as well as the customers who travel to the distant districts.

D2 Operational Burden: D2 gives the highest revenue but serves 95,233 customers (27%) of the customers who are not a part of its natural catchment (D12, D20, D22). Among them, 70,634 customers are within the southwest corridor districts (D12, D22, D24), 20 percent of the total customers of D2. This results in increased unit costs and delivery time, and strain on capacity.

D10 Performance Issues: D10 faces a -25.5% decline and serves only 26.5% market share despite being near high-demand districts. The data indicates that D10 attracts 87,237 customers in D22 and D24 and still loses a huge chunk of share to D2 even in those districts that are geographically nearer to D10.

D14 Efficiency Model: D14 is stable in performance by serving a distinct, local catchment. District 18 has 93% dominance with less competition by D2 or D10, and it has the operational advantages of a clear geographic focus.



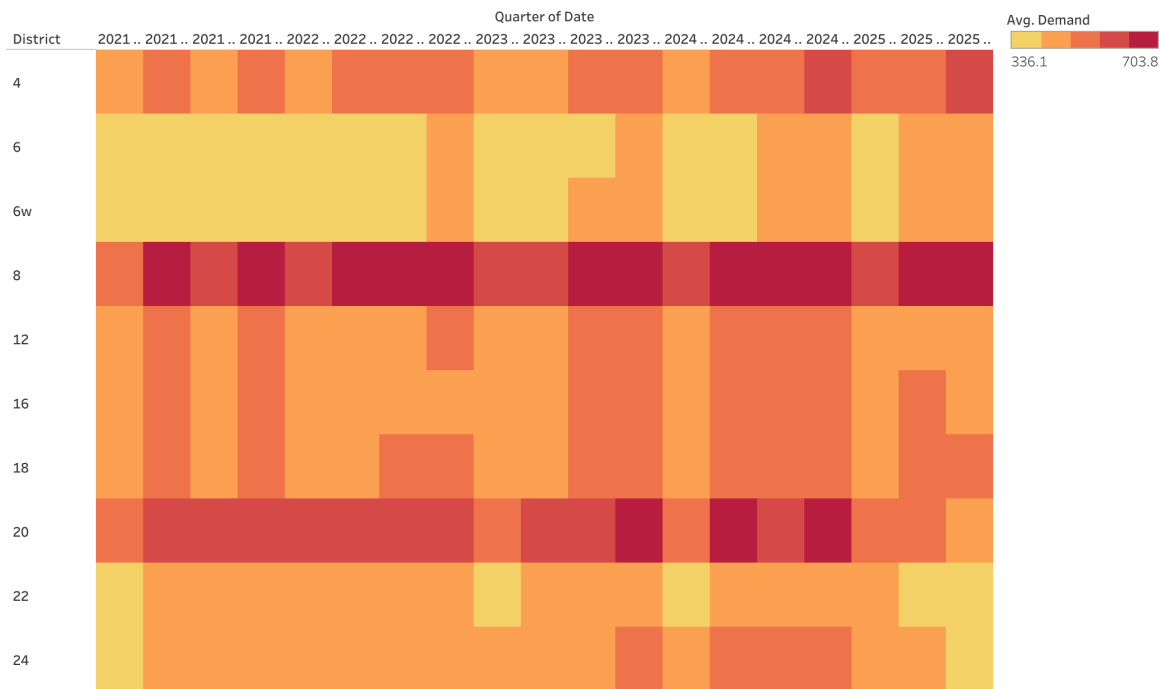
3.2 Coverage Gaps and Demand Concentration

Three areas that show notable patterns relevant to network planning:

1. Southwest Corridor (D12, D22, D24): 235,767 customers who have split coverage in all three branches, and none of them has a dominant local presence.
2. District 16: High D14 presence (80.3% share), which shows it is a well-served neighborhood with little room to grow.
3. District 6w: Moderate D2 presence (62% share) but splitting this catchment would probably decrease its efficiency instead of enhancing it.

The southwest corridor is the only region where the demand (235,767 customers) is high and does not have efficient local service. Customers in these districts are travelling to existing branches, creating operational burden and unpredictability of demand for D2 and D10.

Quarterly Demand Patterns by District (2021-2025)



4. Summary of Findings

Network Performance Overview: The review of the customer demand data (January 2021 - September 2025) shows that there are considerable trends of operational imbalance. Branch D2 dominates the network, serving 27 percent of its large customer base in the remote districts. This reliance on remote customers creates a significant operational burden and a high risk of inefficiency and poor service. Conversely, D14 has a more sustainable model, which serves a localized catchment. In the meantime, D10 is dropping at a rapid rate (-25.5%), grappling with the seeming competitive overlap of the broad scope of D2.

Evidence Base for Decision-Making: This descriptive analysis forms the basis for future expansion and optimization of network decisions. The data clearly shows: (1) presence of operational inefficiency patterns at D2 and D10, (2) the successful, locally oriented service model at D14, and (3) the underserved demand area in the southwest corridor (235,767 customers in D12, D22, D24). These findings make the evidence base for the next phase of analysis and strategic recommendations to the UrbanEats board.

UrbanEats Expansion Analysis

Predictive Analysis for Demand Forecasting – By Satvik Kumar

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Yearly Demand Forecasts (2026-2028)

District	Forecast 2026	Forecast 2027	Forecast 2028
4	20,533	21,455	22,376
6	15,163	15,480	15,798
8	24,086	24,352	24,617
12	17,766	17,932	18,099
16	17,194	17,288	17,383
18	17,969	18,250	18,531
20	20,435	20,222	20,010
22	14,382	12,520	10,658
24	16,042	15,287	14,533
6w	15,253	15,544	15,834

Methodology and Model Selection

Model: Holt-Winters Exponential Smoothing

Holt-Winters Exponential Smoothing was selected as the forecasting method. The preliminary findings showed that the seasonal trends are very strong (73-92% seasonality strength) in all districts, whereas the demand in December is always 30-40% higher than the average. These recurring seasonal trends and underlying growth or decline trends are well represented in the Holt-Winters model.

Model Configuration

Districts were then equipped with an Additive Trend/Additive Seasonal (Add/Add) or Additive Trend/Multiplicative Seasonal (Add/Mul) model depending on the performance of the validation. The model uses three smoothing parameters:

- **Alpha (α):** Determines the sensitivity of the model to the recent changes in the levels of demand. High value is responsive to the new patterns, and low value is used to smooth out the short-term volatility.
- **Beta (β):** Determines the sensitivity to changes in the trend - whether the demand increasing or decreasing with time.
- **Gamma (γ):** Controls how seasonal patterns are updated (lower values preserve stable seasonal patterns)

Parameters were individually optimized per district. For example, District 4: $\alpha=0.1107$, $\beta=0.0722$, $\gamma=0.1531$; District 6w: $\alpha=0.0329$, $\beta=0.0449$, $\gamma=0.1751$. Values ranged from 0.01 to 0.78 across all districts depending on their unique demand characteristics.

Pros: Holt-Winters can simultaneously deal with trend and seasonality, is computationally efficient, generates business forecasts that are interpretable, and can be customized on a district-by-district basis.

Cons: It suffers due to non-uniform patterns of demand, it assumes that the historical patterns will persist, and it needs an adequate amount of data history to create reliable seasonal baselines.

Validation Performance

Validation Approach: The 57-month data (January 2021 - September 2025) was split into training (45 months: Jan 2021 - Sep 2024), and testing (12 months: Oct 2024 - Sep 2025). Historical patterns were used to train the models and validate them using the 12-month test period and predicting the future demand against actual demand.

Error Metrics: The accuracy of the forecasts was measured by the Mean Absolute Percentage Error (MAPE). The average MAPE was **6.66%**, which implies that it has a good predictive power.

Performance breakdown:

- **High accuracy districts** (MAPE 2.73%-4.87%): Districts 4, 6, 8, 12, 16, 18, 6w
- **Moderate accuracy districts** (MAPE 8.92%-16.76%): Districts 20, 22, 24

Increased errors in Districts 20 and 22 represent truly volatile historical trends with irregular swings that are, in themselves, hard to forecast.

Key Observations

- **Growth Trajectory:** The forecast indicates that seven districts have a stable growth of 1-5% per year.
- **Declining Markets:** Two districts, 22 and 24, are expected to experience a falling demand. District 22 is a major issue, and its demand is expected to decrease by 26% in 2028.
- **Stabilizing Market:** District 20 seems to be stabilizing, with the demand levelling out around 20,000 orders per year.
- **Forecast Uncertainty:** The forecasts for Districts 20 and 22 have wider confidence intervals. This is because they have irregular and volatile historical trends and are therefore more difficult to predict.

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Prescriptive Analysis for Expansion Recommendation – By Satvik Kumar

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Optimization model recommends opening 5 new branches in Districts D8, D12, D16, D20, D24 to maximize customer demand (95,082 annual average 2026–2028) while meeting budget and distance constraints. This addresses Part 1 findings: D2 operational burden serving 27% remote customers, southwestern corridor gap with 235,767 underserved customers, and validates Part 2 demand forecasts (6.66% MAPE).

Primary Recommendation

Metric	Value	Status
Average 2026–2028 Demand	95,082 customers	Optimal
Total Investment	€1,360,000	€220K under budget
Average Distance	22	Exceeds 12.9 requirement
Selected Branches	D8, D12, D16, D20, D24	5 locations

Selected Branch Locations

District	Cost	2026 Demand	Avg 2026–2028	Rationale
D8	€480K	24,086	24,352	Highest demand; stable growth
D12	€150K	17,766	17,932	Lowest cost; best ROI
D16	€350K	17,194	17,288	Reliable; moderate investment
D20	€200K	20,435	20,222	High demand; cost efficient
D24	€180K	16,042	15,287	SW corridor completion

Excluded: D4 (€1,600K prohibitive), D22 (–26% decline), D6/D6w (lower priority), D18 (€875K high cost).

Alternative Scenarios

Scenario	Districts	Demand	Key Insight
Budget-Constrained (€1.2M)	D12, D16, D20, D22, D24	83,250	–12% demand vs base; forces risky D22
Operational Focus (Distance ≤ 15)	D6w, D8, D12, D20	78,050	–18% demand; tighter geographic cluster
Conservative (≤ 3 Branches)	D8, D18, D20	62,824	–34% demand; quality-over-quantity approach

Conclusion

The 5-branch recommendation (€1,360,000, 95,082 annual demand) maximizes customer reach while addressing Part 1 analysis and validating Part 2 forecasts. Opens D8 & D20 immediately for rapid ROI; stagger D12, D16, D24 pending demand validation. Maintains €220K contingency and operational flexibility.